YANG ZHANG

Phone number: +86 18322749801 | Email: yangz@mail.nankai.edu.cn | WeChat: -zzzz123

EDUCATION

2025 - Present	Nankai University, College of Cyber Science and Engineering | M.S. in Cyberspace Security Advisor: Prof. Zheli Liu; research focus: vector databases, AI security, encrypted database security	Tianjin, China
2021 - 2025	Nankai University, College of Cyber Science and Engineering | B.E. in Information Security Rank: 7/50; GPA: 3.7/4.0; Weighted Score: 89.01/100	Tianjin, China

RESEARCH INTERESTS AND SKILLS

Research interests: AI security, RAG privacy, vector database leakage, encrypted database security, searchable encryption, security evaluation.

Programming and systems: C/C++ (multithreading, memory management), Python (experiments and automation), Java; Git, Docker, Linux, openGauss, SQL.

Security: SSE/StE leakage analysis, RAG extraction attacks and defenses, vulnerability reproduction, POC development, Web security testing.

PUBLICATIONS AND MANUSCRIPTS

SLIP Through the Cracks: Semantic Leakage from ID Patterns in Vector Database Similarity Queries | Under submission to NDSS | First Author

- Identified a semantic leakage channel where an observer can infer query labels from top-k returned ID lists without seeing query vectors or database coordinates.
- Built a three-stage pipeline: shadow graph construction from ID-list co-occurrence, spectral clustering for unlabeled semantic groups, and label assignment with class-size priors or anchor queries.
- Evaluated the attack on text and image datasets; observed intra/inter Jaccard ratios from 5.2x to 40.9x and analyzed defenses including rank shuffling, decoy IDs, and smaller retrieval depth.

Can I Get More? An Incremental Inference Attack on Encrypted SQL | IEEE Symposium on Security and Privacy (S&P/Oakland) 2026 | Second Author

- Contributed to algorithm design, implementation, and experiments for Anchor-Joint Attack, an incremental plaintext recovery attack against encrypted SQL leakage.
- Implemented frequency inference via Optimal Transport, using EMD/Sinkhorn-based ciphertext-plaintext matching and confidence-driven anchor selection.
- Ran experiments on healthcare and urban datasets against Single, Jigsaw, Greedy, Genetic, and Split baselines; the method achieved up to 71.68% value recovery and 95.45% row recovery.

"You Shouldn't Have Taken It": Detecting Extraction from Retrieval-Augmented Generation via Hidden Representations | Under submission to KDD | Co-first Author

- Worked on algorithms, code, and experiments for DIVER, a pre-generation detector for external data extraction attacks in RAG systems.
- Extracted hidden states from frozen LLMs at the last query token and trained lightweight probes to identify extraction intent before any token is generated.
- Evaluated query-only training and in-RAG inference across multiple backbones, including Qwen2.5, Vicuna, Llama2, and Mistral; ranked 22nd in the DataCon2025 AI Security track.

SELECTED ENGINEERING AND SECURITY PROJECTS

Dynamic Searchable Encryption Prototype in openGauss | C++ / SQL / openGauss / libpq

- Integrated a Fides/Sophos-style dynamic searchable encryption prototype into openGauss with custom commands including Fides_Insert, Fides_Select, and Fides_Delete.
- Modified the libpq query submission path to rewrite Fides SQL into encrypted-index updates, search token generation, and underlying SQL execution.
- Implemented client-side keyword-counter state management, token derivation, deletion filtering, and a two-stage encrypted-index lookup plus data-table retrieval flow.

Security R&D / Vulnerability Analysis Practice, 360 Enterprise Security Group | Python / Web Security / POC

- Reproduced and analyzed high-risk vulnerabilities, including Apache bRPC RCE and VMware vCenter Server RCE (CVE-2021-21972); documented triggering conditions, exploitation paths, and remediation advice.
- Conducted Web security testing for SQL injection, file upload, file parsing, information disclosure, and WAF log analysis; wrote Python POC scripts and validation procedures.

Large-Model-Assisted College Application Recommendation System | Nankai University President Cup

- Developed an intelligent application for college-major recommendation, responsible for front-end, back-end, database design, and LLM API integration.

HONORS AND AWARDS

DataCon2025 AI Security Track: 22nd among 541 teams; **National Internet+ Innovation and Entrepreneurship Competition:** Silver Award; **MCM/ICM 2024:** Honorable Mention; **Nankai University:** Outstanding Graduate, Academic Excellence Scholarship, Gongneng Scholarship, Merit Student.